1. Find Duplicate Elements in an Array

10 marks

Write a Java program to find all duplicate elements in an integer array without using collections or built-in sorting methods. Print each duplicate only once.

Input

4 2 7 4 8 2 9 4

Expected output

Duplicate elements: 4 2

Concepts: Arrays, nested loops, conditions.

2. Sort an Array Without Built-in Methods

10 marks

Sort an integer array in ascending order using a sorting algorithm such as Bubble Sort. Do not use Arrays.sort().

Input

35 12 8 67 21 4

Expected output

4 8 12 21 35 67

Concepts: Arrays, nested loops, swapping.

3. Check Whether Two Strings Are Anagrams

10 marks

Determine whether two strings contain the same characters with the same frequencies, ignoring case and spaces. Do not use Arrays.sort() or collections.

Input

Dormitory

Dirty room

Expected output

The strings are anagrams.

Concepts: Strings, character arrays, loops, frequency counting.

4. Find the First Non-Repeating Character

10 marks

Given a string, find the first character that appears exactly once. If no such character exists, display an appropriate message.

Input

swiss

Expected output

First non-repeating character: w

Concepts: Strings, character frequency, nested loops.

5. Rotate an Array by K Positions

10 marks

Rotate an array to the right by K positions. Handle cases where K is greater than the array length.

Input

Array: 1 2 3 4 5 6 7

K: 3

Expected output

5 6 7 1 2 3 4

Concepts: Arrays, loops, index calculations.

6. Build a Student Result Management System

10 marks

Create a Student class with roll number, name, and marks in five subjects. Calculate total, average, and grade. Include at least three student objects and display the topper.

Rules

- Average ≥ 90: Grade A
- Average ≥ 75: Grade B
- Average ≥ 60: Grade C
- Average ≥ 40: Grade D
- Otherwise: Grade F

Expected output example

Name: Arun

Total: 425

Average: 85.0

Grade: B

Topper: Priya

Concepts: Classes, objects, arrays, constructors, methods.

7. Implement a Shape Area Calculator Using Polymorphism

10 marks

Create a parent class Shape with an area() method. Override it in Circle, Rectangle, and Triangle. Store the objects in a Shape array and display each area's result using polymorphism.

Sample dimensions: Circle radius 7, rectangle 5 × 4, triangle base 6 and height 8.

Expected output

Circle area: 153.94

Rectangle area: 20.0

Triangle area: 24.0

Concepts: Inheritance, overriding, abstraction or polymorphism.

8. Create a Custom Exception for Bank Withdrawals

10 marks

Create a BankAccount class with a private balance. Implement deposit and withdrawal methods. Create a custom InsufficientBalanceException and throw it when a withdrawal exceeds the available balance.

Example

Initial balance: 5000

Withdraw: 6000

Expected output

Insufficient balance!

Available balance: 5000

Concepts: Encapsulation, custom exceptions, throw, throws, try-catch.